

Assignment #1: Data Science (Spring 2026)

Experimental Study with WEKA

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Question 1: Experimental Study with WEKA

1.1 1. Datasets Selection

For this experimental study, three datasets were selected from the Kaggle and UCI Machine Learning repositories. These datasets represent diverse domains (Software Engineering, Botany, and Health) to ensure a robust evaluation of the classifiers.

- **Bug Reports (IT / Software):** A dataset containing software bug metadata (e.g., error code, tech stack) used to classify the type of bug (e.g., Memory Leak, API Bug). Target Variable: *Bug Category*.
- **Iris (Botany):** A classic dataset predicting the species of Iris flowers (*Setosa*, *Versicolor*, *Virginica*) based on sepal and petal measurements. Target Variable: *Species*.
- **Diabetes (Health):** Medical data used to predict whether a patient has diabetes (Positive/Negative) based on diagnostic metrics like Glucose, BMI, and Age. Target Variable: *Outcome*.

1.2 2. Algorithms Selected

Three distinct classification algorithms were applied to compare their performance:

1. **J48 (Decision Tree):** An implementation of the C4.5 algorithm that builds a decision tree by splitting data based on information gain, creating interpretable "if-then" rules.
2. **Naive Bayes:** A probabilistic classifier based on Bayes' theorem. It assumes that features are independent of each other, making it fast and effective for high-dimensional data.
3. **Random Forest:** An ensemble learning method that constructs multiple decision trees during training and outputs the mode of the classes. It is generally robust against overfitting.

1.3 3. Experimental Setup

The experiments were conducted using the **WEKA Experimenter** interface.

- **Validation Method:** 10-fold Cross-Validation.
- **Process:** The data was divided into 10 parts, training on 9 and testing on 1, repeating the process 10 times to ensure statistical reliability.

1.4 4. Results and Analysis

A. Accuracy Comparison (Percent Correct)

Table 1 shows the percentage of correctly classified instances for each algorithm across the three datasets.

Dataset	J48 (Decision Tree)	Naive Bayes	Random Forest
Bug Reports	100.00 %	100.00 %	100.00 %
Iris	94.73 %	95.53 %	94.67 %
Diabetes	65.11 %	67.62 %	64.05 %

Table 1: Comparison of Algorithm Accuracy

B. Weighted Recall Comparison

Table 2 compares the Weighted Recall, which measures the ability of the classifier to identify positive samples.

Dataset	J48 (Decision Tree)	Naive Bayes	Random Forest
Bug Reports	1.00	1.00	1.00
Iris	0.95	0.96	0.95
Diabetes	0.65	0.68	0.64

Table 2: Comparison of Weighted Recall

C. Weighted Precision Comparison

Table 3 presents the Weighted Precision for each algorithm.

Dataset	J48	Naive Bayes	Random Forest
Bug Reports	1.00	1.00	1.00
Iris	0.96	0.96	0.96
Diabetes	0.66	0.69	0.65

Table 3: Comparison of Weighted Precision

Analysis of Precision

- **Bug Reports:** All algorithms achieved a perfect precision score of **1.00**, confirming that the model made no false positive errors.
- **Iris:** Precision was consistently high (**0.96**) across all classifiers, meaning that when the model predicted a flower species, it was correct 96% of the time.
- **Diabetes:** Precision was lower, ranging from **0.65 to 0.69**. Naive Bayes achieved the highest precision (0.69), suggesting it is slightly better at distinguishing between diabetic and non-diabetic patients compared to J48 and Random Forest, which had more false positives.

D. Graphical Analysis

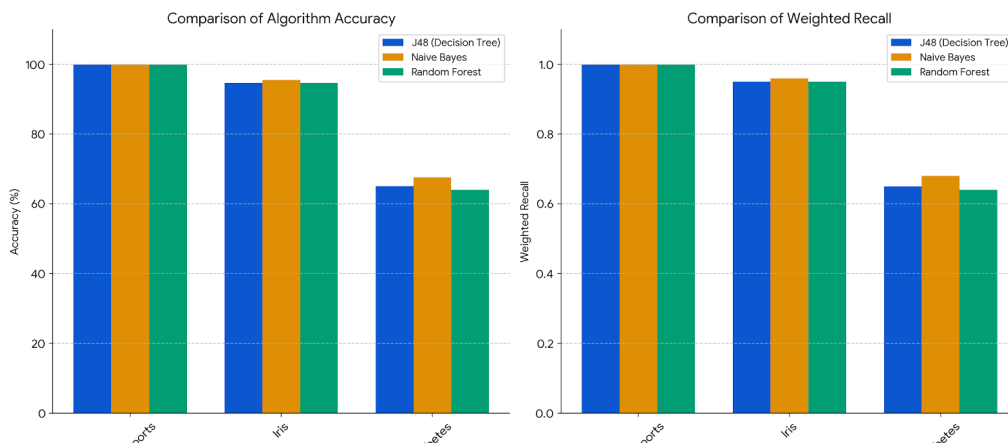


Figure 1: Bar Chart comparing Accuracy and Recall across algorithms.

1.5 5. Critical Discussion

Performance on Bug Reports (100% Accuracy): All three algorithms achieved perfect accuracy on the Bug Reports dataset. This is an unusual result in real-world data and suggests that the dataset likely contains a "giveaway" feature—such as an Error Code that maps directly to the Bug Category. In a real-world scenario, such highly correlated features would be removed to force the model to learn from more subtle patterns.

Performance on Iris (Naive Bayes Wins - 95.53%): Naive Bayes slightly outperformed J48 and Random Forest on the Iris dataset. The Iris dataset features (petal/sepal lengths) follow a Gaussian (normal) distribution, which aligns perfectly with the mathematical assumptions of the Naive Bayes classifier. J48 also performed well because the decision boundaries in Iris are simple and rectangular.

Performance on Diabetes (Naive Bayes Wins - 67.62%): The Diabetes dataset was the most challenging. Naive Bayes performed the best (67.62%), significantly outperforming J48 and Random Forest. This suggests that the relationship between health metrics (like Glucose and Age) and Diabetes is probabilistic rather than a strict set of rules. Random Forest likely "overfitted" the noisy data, leading to lower accuracy compared to the simpler Naive Bayes model.